

RENATO LA TORRE
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EDUCATION

- 2023 – 2025 **MSc. Biology – Ecology, Behavior, Evolution and Biosystematics, Norwegian University of Science and Technology (NTNU - Trondheim, Norway)**
Main subject studies: ecology, statistics, population genetics, conservation.
Thesis grade: A
- 2015 – 2020 **B.S. Genetics and Biotechnology, Universidad Nacional Mayor de San Marcos (UNMSM - Lima, Peru)**
Main subject studies: genetics, biotechnology, molecular biology, bioinformatics.
Percentile: Top 15%
Honors: Summa Cum Laude

SKILLS

Research

- Bioinformatics: WGS data analysis for (meta)barcoding, genome assembly and population genomics, ancient DNA. Knowledge of R, Python and Bash. Data and script archiving.
- Molecular biology: Execution and troubleshooting of DNA/RNA extraction methods, PCR/qPCR, Illumina/ONT sequencing library preparation. Modern and ancient samples.

Supervision and teaching

- Experience in participating and leading interdisciplinary groups.
- Training students on laboratory techniques and bioinformatic pipelines.
- Teaching assistant in Bioinformatics, Phylogeography and Population Genetics courses.

Other professional activities

- Excellent oral presenter.
- Fieldwork experience in natural and archaeological settings.
- Good lab worker– efficient, organized, disciplined, and detail-oriented.
- Member of conservation and science-promoting programs.

WORK EXPERIENCE

- Aug 2025 – Present Lab. Genomics and Bioinformatics for Biodiversity (LGBB) – UNMSM Bioinformatician – Responsible for the laboratory's computing resources and bioinformatics training.
- Nov 2024 – May 2025 Institutt for naturhistorie (INH) – NTNU Vitenskapsmuseet
Student assistant – Collaboration in projects in ancient genomics.
- Jan 2021 – May 2023 Lab. Genomics and Bioinformatics for Biodiversity (LGBB) - UNMSM
Research assistant – Collaboration in projects related to genomics and metabarcoding of endangered tree *Neltuma pallida*.

Jan 2022 – Apr 2022	<u>Technological, Biomedical and Environmental Research Center (CITBM)</u> <i>Teaching assistant</i> – Logistics for the “International Course of Bioinformatics and Genomics for the Biodiversity”.
Jul 2020 – Dec 2020	<u>Tradupharma EIRL</u> <i>Translator</i> - Translation of scientific literature.

TRAINING EXPERIENCE

Aug 2025	Workshop, “Summer School Introduction to Ancient Metagenomics” SPAAM (online), Aug 4 th – 8 th , 2025.
Jan 2025	Course, “Recent Advances in Conservation Genetics, ConGen 2025” Cheetah Conservation Fund Research Center, Namibia, Jan 1 st – 15 th , 2025.
Aug 2024	Course, “Barcoding with ONT MinION: Theory and Practice” NTNU, Norway, Aug 19 th – 22 nd , 2024.
Mar 2024	Course, “BIO9001: Population Genomics Nord 2024” Nord University, Norway, March 18 th – 22 nd , 2024.
Oct 2023	Course, “MOL8008: Bioinformatics Methods for NGS Analysis” NTNU, Norway, Oct 2 nd – 6 th , 2023.
May 2023	Course, “Introduction to Palaeogenomics” Transmitting Science, Peru, May 15 th – 26 th , 2023.
Mar 2023	Workshop, “RNA and transcriptome analysis for organismic biology” Museo de Historia Natural, Peru, Mar 8 th – 11 th , 2023.
Nov 2020	Course, “Introduction to Bacteria Phylogenomics” BioScience-App Courses, Peru, Oct 13 th to Nov 3 rd , 2020.
Jan 2020 – Mar 2020	Research Internship at Texas A&M University Symbiosis Lab (PI: Dr. Mariana Mateos), Department of Wildlife and Fisheries Sciences, Texas A&M University, College Station, Texas, United States of America, Jan 13 th to Mar 30 th , 2020.
Jan 2018 – Jan 2020	Assistantship, Lab. Molecular Ecology and Aquatic Biodiversity Responsible: MSc. Daniel Ore, Biological Sciences Faculty, San Marcos University, Lima, Peru.
Nov 2017 – Oct 2019	Assistantship, Lab. Genetics of Metabolic Diseases Responsible: Dr. Monica Paredes, Biological Sciences Faculty, San Marcos University, Lima, Peru.
Sep 2019	Workshop, “GWAS, post-GWAS, WGS, WES and population genetics” Centro de Convenciones Daniel Alcides Carrión, International School of Hereditary Ataxias: from genetics to the clinic – ISIA 2019, Lima, Peru, Sep 17 th – 19 th , 2019.
Jan 2019 – Mar 2019	Internship, “Academic Rotation on Molecular Neurogenetics” Center for Basic Research in Neurogenetics, National Institute for Neurologic Sciences, Lima, Peru, Jan 1 st to Mar 29 th , 2019.
Sep 2018	Course, “Using Next Generation Sequencing to Study Microbial Genetic Diversity” XXII International Congress of Genetics, Foz do Iguaçu, Brazil, Sep 10 th , 2018.

- Jun 2018 **Program, “Students Leaders of the 21st Century”**
Harvard University, Cambridge, Massachusetts, United States of America, Jun 4th – 15th, 2018.
- Jan 2018 **Course, “Introduction to Phylogenetics”**
Institute of Genetics Barbara McClintock, Peru, Jan 18th – 20th, 2018.
- Sep 2017 – Oct 2017 **Course, “SNPs Analysis using Real-Time PCR and High-Resolution Melting”**
Institute of Genetics Barbara McClintock, Peru, Sep 25th to Oct 1st, 2017.
- Aug 2017 – Sep 2017 **Workshop, “Applied Bioinformatics”**
Peruvian Institute of Genetics (IPEGEN), Peru, Aug 24th to Sep 7th, 2017.
- Jul 2017 **Workshop, “Clubes de Ciencia Perú (Science Club Peru)”**
Universidad Peruana Cayetano Heredia, Peru, Jul 22th – 27th, 2017.

SCIENTIFIC ARTICLES

- Jul 2025 **Preprint, “Haplotype-resolved chromosome-level genome assemblies of four *Diamesa* species reveal the genetic basis of cold tolerance and high-altitude adaptations in arctic chironomids”**
Authors: Martin S.L.F., **La Torre R.**, Danneels B., Tooming-Klunderud A., Skage M., Kollias S., Tørresen O.L., Anbaran M.F., Stur E., Jaboksen K.S., Martin M.D., Ekrem T.
bioRxiv, 2025-07
<https://doi.org/10.1101/2025.07.30.667619>
- Jun 2025 **Preprint, “Sedimentary ancient DNA metagenomic analysis provides new insights into farming in central Norway from the Bronze Age to late Medieval period”**
Authors: **La Torre R.**, Kosel P., Bieker V., Westergaard K.B., Mykkeltvedt J.L., Yildirim K.L., Seiler M., Philippsen B., Macphail R.I., Cannell R.J.S., Halvorsen L.S., Lorentzen A.B., Sauvage R., Martin M.D., Breivik H.M., Martin S.L.F.
bioRxiv, 2025-06
<https://doi.org/10.1101/2025.06.19.660509>
- May 2025 **“Lima megacity’s influence on aquatic microbial communities in the Rímac River: Dominance over spatial and seasonal variations”**
Authors: Samaniego T., **La Torre R.**, Orjeda G., Ramirez J.L.
Microbial Ecology, 88(1), 57.
<https://doi.org/10.1007/s00248-025-02558-5>
- Feb 2025 **“A chromosome-level genome assembly of the Peruvian Algarrobo (*Neltuma pallida*) provides insights on its adaptation to its unique ecological niche”**
Authors: **La Torre R.**, Hamilton J.P., Saucedo-Bazalar M., Caycho E., Vaillancourt B., Wood J.C., Ramirez M., Buell C.R., Orjeda G.
G3: Genes, Genomes, Genetics, 15.2 (2025), jkae283
<https://doi.org/10.1093/g3journal/jkae283>
- Nov 2023 **“Assembly, annotation and analysis of the chloroplast genome of the Algarrobo tree *Neltuma pallida* (subfamily: Caesalpinioideae)”**
Authors: Caycho E., **La Torre R.**, Orjeda G.
BMC Plant Biology, 23.1 (2023): 570
<https://doi.org/10.1186/s12870-023-04581-5>
- May 2023 **“Survey of lichenized-fungi DNA barcodes in King George Island (Antarctica): An aid to species discovery”**

Authors: **La Torre R.**, Ramos D., Mejía M., Neyra E., Loarte E., Orjeda G.
Journal of Fungi (Switzerland), 9.5 (2023): 552
<https://doi.org/10.3390/jof9050552>

- Dec 2019 **“Association analysis between *PTPN1* gene polymorphisms (rs941798 and rs914458) and diabetes type II in families from Lima-Peru”**
Authors: Sanchez E., **La Torre R.**, Garcia-Quispes W., Paredes-Anaya M.
Horizonte Médico (Lima) 19.4 (2019): 14-19.
<http://dx.doi.org/10.24265/horizmed.2019.v19n4.03>

CONFERENCE PRESENTATIONS

- Dec 2022 **“Population genomics study of algarrobo *Prosopis pallida* (=Neltuma pallida)”**
Authors: **La Torre R.**, Caycho, E., Ñaupari, K., Suarez, R., Orjeda, G.
I International Symposium on Plant Genomics and Bioinformatics, in Lima, Peru.
- Jul 2022 **1. “Evaluation of the identification of lichenized fungi from King George’s Island, Antarctica, using DNA barcoding”**
2. “Standardization of a methodology for DNA extraction and amplification of lichenized fungi from King George Island, Antarctica”
Authors: **La Torre R.**, Mejía M., Loarte E., Cotera C., Neyra E., Orjeda G.
XV Lichenology Latin American Group (GLAL) Meeting, in Argentina.
- Apr 2020 **“Detection of viral RNA sequences in transcriptomes from three species of *Prosopis* (Fabaceae) published in NCBI”**
Authors: **La Torre R.**, Rivera, A., Lozano, R., Orjeda, G.
I National Congress of Life Sciences and XXVII ICBAR Scientific Meeting, in Lima, Peru.
- Apr 2020 **“Effect of wasp parasitism on the bacterial densities in the *Spiroplasma-Drosophila melanogaster* symbiosis”**
Authors: **La Torre R.**, Higareda V., Stankov A., Martinez-Romero E., Mateos M.
XXI Ecological Integration Symposium, in College Station, Texas, United States of America.
- Jan 2019 **“In silico EPIC-marker identification in the *Pomacea* genus”**
Authors: **La Torre R.**, Solís M., Silvestre T., Oré D.
IV International Scientific Journey of Biosciences, in Lima, Peru.
- Sep 2018 **“Family-based association study between SNPs rs914458 and rs941798 (*PTPN1* gene) with type 2 diabetes mellitus”**
Authors: **La Torre R.**, Torres D., Padilla C., García-Quispes W., Calderón J., Paredes Anaya M.
XXII International Congress of Genetics, in Foz do Iguaçu, Paraná, Brazil.
- Dec 2017 **“Primer design of conserved microsatellite markers using taxon Gekkota expression sequence tags”**
Authors: **La Torre R.**, Velásquez A., Ojanama B., Justino S., Ccama S., Toledo G., Oré D.
III International Scientific Journey of Biosciences, in Lima, Peru.

SCHOLARSHIPS AND RECOGNITIONS

- Sep 2024 **Admission: “ORG.one program”**
Source: Oxford Nanopore Technologies
Support of consumables for the whole-genome sequencing of endangered species *Telmatobius macrostomus*.
- Aug 2024 **Scholarship: “Partial Fund for Conservation Genetics 2025” (1,250 USD)**

Source: The American Genetic Association (AGA)
Support for registration and expenses for attending the ConGen 2025 course.

- Jun 2024 **Grant: “Sparebanken Midt-Norges gavfond til Norges teknisk-naturvitenskapelige universitet” (1,909.78 USD)**
Source: UNIFOR
My project titled “Unveiling the genetic diversity of Peruvian mesquite (*Neltuma pallida*) to illuminate its exceptional adaptation to arid environments” was selected for funding.
- Dec 2022 **Fellowship: “Graduate student training in biodiversity genomics (BiGTREE)” (36,762.50 USD)**
Source: Norges teknisk-naturvitenskapelige universitet (NTNU)
Support for training and living expenses during master degree studies at NTNU in Trondheim, Norway.
- Jun 2022 **Recognition: “Thesis defense approved with the highest distinction”**
Source: Universidad Nacional Mayor de San Marcos (UNMSM)
My thesis titled “Standardization and evaluation of a methodology based on DNA barcoding for the identification of lichens from King George Island, Antarctica” was approved with the honorary title “Summa cum laude”.
- May 2022 **Grant: “Fund for short-term research projects” (1,328.73 USD)**
Source: National Service of Natural Areas Protected by the State (SERNANP)
My project titled “Population genomics of *Prosopis pallida* through genotype by sequencing” was selected for funding.
- Apr 2022 **Admission: “Opportunity Funds Program”**
Source: U.S. Department of State Bureau of Educational and Cultural Affairs (EducationUSA)
Support of preparing, applying and/or beginning studies at U.S. higher education institutions.
- Oct 2019 **Fellowship: “Funding for research thesis project” (2,125.96 USD)**
Source: National Institute for Research on Glaciers and Mountain Ecosystems (INAIGEM)
My project titled “DNA barcoding of Antarctic lichens” was selected for funding.
- Aug 2019 **Admission: “Research Experience for Peruvian Undergraduates (REPU)”**
Training and mentorship for international internship at Texas A&M. Given to outstanding Peruvian students.
- Jan 2019 **Recognition: “IV International Scientific Journey of Biosciences”**
My poster titled “*In silico* EPIC-marker identification in the *Pomacea* genus” was highlighted at the event.
- Sep 2018 **Scholarship: “Permanencia Académica” (3,188.94 USD)**
Source: National Program of Scholarships and Educational Loan
Support for living expenses for outstanding students with limited resources.
- May 2018 **Admission: “Students Leaders of the 21st Century”**
Source: Laspau (US) and Universidad Nacional Mayor de San Marcos (UNMSM)

Training on leadership and other soft skills at Harvard University

Dec 2017 **Recognition: “III International Scientific Journey of Biosciences”**
My poster titled “Primer design of conserved microsatellite markers using taxon Gekkota ESTs” was highlighted at the event.

VOLUNTARY WORK

Jun 2025 – Present	Participant of cleaning and environmental outreach campaigns led by the NGO “Perú Te Quiero Limpio”.
Jul 2023	Member of the Selection Committee in the Ecology branch for REPU, a program supporting international internships for outstanding students.
Jul 2021	Member of the Logistics Committee in the I National Congress of Life Sciences and XXVII ICBAR Scientific Meeting.
May 2021 – Jan 2022	Teaching assistant in Population Genetics 2021-I and 2021-II (Prof. Gisella Orjeda) at San Marcos University (UNMSM-Peru).
Jan 2021 – Apr 2021	Member of “REPU Buddies” , program aiming to give friendly advice to newcomers to the REPU program.
Nov 2020 – Jan 2021	Moderator for the Genomics Journal Club at San Marcos University (UNMSM-Peru)
May 2020 – Nov 2020	Teaching assistant in Phylogeography (Prof. Rina Ramirez) and Population Genetics (Prof. Gisella Orjeda) at San Marcos University.
Feb 2020	Participant in Darwin Day 2020, a science outreach event at Texas A&M University.
Nov 2018	Scientific expedition “Looking for the threatened frogs of Junin and Pasco, <i>Telmatobius macrostomus</i> and <i>T. brachydactylus</i> ” with RANA, a conservation non-governmental organization.
Jul 2018 – Aug 2018	Member of the logistics of “Clubes de Ciencia Perú”, a science outreach event and workshop for high school and early college students.
Feb 2018 – May 2018	Assistant in science outreach activities promoted by RANA, to raise awareness about <i>Phyllodactylus sentosus</i> and <i>Telmatobius macrostomus</i> , two endangered endemic species.
Jan 2017 – Oct 2017	Volunteer in cleaning campaigns organized by the Peruvian Ministry of Environment.

REFERENCES

Prof. Gisella Orjeda
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